
Tameness, Maximal Immediate Extensions, and the Decidability Boundary

A Criterion, a Transfer Theorem, and a Sharp Example

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Abstract

A maximal immediate extension of a valued field is a canonical completion-like object: it always exists, and it is henselian, complete and defectless. It need not be unique, however, and its first-order theory in the language of valued rings can be far richer than that of the field it extends. This paper isolates exactly when that theory is controlled. Working from Kuhlmann's characterisation of tame fields, we prove that a maximal immediate extension is tame if and only if the base satisfies two elementary conditions on its value group and residue field, and we deduce that tameness is a permanence property: it is inherited by passing to a maximal immediate extension. Combining this with relative completeness and relative decidability for tame fields yields a decidability transfer theorem in equal characteristic, and its contrapositive gives a usable obstruction: undecidability of a maximal immediate extension certifies that the base is not tame. We exhibit the sharp phenomenon at the boundary, a maximal immediate extension whose valued-field theory is decidable while its expansion by a named element is undecidable, obtained from the tilting correspondence. We then draw out applications to effective certification, invariant transport, and the classification of candidate fields.

1 Introduction

Let (K, v) be a valued field. Its maximal immediate extension (Z, w) , maximal among immediate extensions of (K, v) , exists by a theorem of Krull [1], simplified by Gravett [2]. It is unique up to valuation-preserving isomorphism over K when (K, v) is a Kaplansky field [3], but not in general; Section 6 records a case in which pairwise nonisomorphic maximal immediate extensions occur. It does enjoy strong structural properties: it is henselian, complete and defectless [4, Thm. 2.12]. It is therefore a natural object. The question we address is when its first-order theory is *decidable*, since decidability of $\text{Th}(Z, w)$ determines what can be algorithmically verified about it.

The naive expectation is that decidability should be governed by tameness of the base field. This turns out to be too coarse. The precise invariant is narrower: tameness of (Z, w) depends only on two readily checkable conditions on the value group and residue field of (K, v) , and not on whether (K, v) is itself maximal or algebraically maximal. Once tameness of (Z, w) is established, relative decidability for tame fields takes over, and in equal characteristic the entire question reduces to the elementary theories of two much simpler objects.

Our contributions are these.

1. An exact criterion: (Z, w) is tame if and only if (K, v) satisfies (TF1) and (TF2) (Theorem 3.2).
In particular tame fields are closed under maximal immediate extension.
2. A decidability transfer theorem for equal characteristic (Theorem 4.2).
3. Its contrapositive, usable as a negative certificate (Corollary 4.3).
4. A sharp example at which tame theory stops controlling things (Theorem 5.2), in which $\text{Th}(Z, v)$ is decidable but $\text{Th}(Z, t)$ is not.
5. Applications, in Section 7.

2 Notation and definitions

Throughout (K, v) is a valued field, $\Gamma_v = v(K^\times)$ its value group, $\kappa(v) = Kv$ its residue field, $\mathcal{O}_v$ its valuation ring and $\mathfrak{m}_v$ its maximal ideal. Write L_{val} for the language of valued rings and $L_{\text{val}}(t)$ for L_{val} with a distinguished constant t .

An extension (L, w) of (K, v) is *immediate* if $\Gamma_w \cong \Gamma_v$ and $\kappa(w) \cong \kappa(v)$. A valued field is *algebraically maximal* if it admits no proper immediate algebraic extension, and *maximal* if it admits no proper immediate extension. It is *defectless* [7] if every finite extension M of K satisfies $[M : K] = (\Gamma_M : \Gamma_v)[\kappa(M) : \kappa(v)]$. The field is *henselian* if v extends uniquely to every algebraic extension.

Definition 2.1. For a henselian valued field (K, v) consider the conditions

- (TF1) if $\text{char } \kappa(v) = p > 0$, then Γ_v is p -divisible,
- (TF2) $\kappa(v)$ is perfect,
- (TF3) (K, v) is algebraically maximal.

Theorem 2.2 (Kuhlmann's characterisation). *A henselian valued field is tame if and only if it satisfies (TF1), (TF2) and (TF3) [8, Thm. 8].*

Theorem 2.3 (Maximal fields). *A maximal valued field is henselian, complete and defectless [4, Thm. 2.12].*

Theorem 2.3 is the structural engine of the paper: it is what lets us discharge the henselian part of Theorem 2.2 for a maximal immediate extension without further work.

3 The criterion

Lemma 3.1 (Invariants are inherited by immediacy). *Let (L, w) be an immediate extension of (K, v) . Then $\Gamma_w \cong \Gamma_v$ and $\kappa(w) \cong \kappa(v)$. Consequently: if $\text{char } \kappa(v) = p > 0$ then $\text{char } \kappa(w) = p$; the condition " Γ is p -divisible" is equivalent for Γ_v and Γ_w ; and " κ is perfect" is equivalent for $\kappa(v)$ and $\kappa(w)$.*

Proof. Both identifications are precisely the definition of immediacy. The three conclusions are the transitivity of group isomorphism, of field isomorphism, and the invariance of p -divisibility and perfectness under isomorphism. $\square$

Theorem 3.2 (Exact tameness criterion). *Let (Z, w) be a maximal immediate extension of (K, v) . Then (Z, w) is tame if and only if (K, v) satisfies (TF1) and (TF2).*

Proof. Suppose first that (K, v) satisfies (TF1) and (TF2). By Lemma 3.1 the same two conditions hold for (Z, w) . A maximal field is algebraically maximal: an immediate algebraic extension would be an immediate extension, contradicting maximality. So (TF3) holds for (Z, w) . By Theorem 2.3, (Z, w) is henselian. Theorem 2.2 now gives tameness.

Conversely suppose (Z, w) is tame, so it satisfies (TF1) and (TF2) by Theorem 2.2. By Lemma 3.1 both transfer back to (K, v) . $\square$

Corollary 3.3. *Tameness is a permanence property: if (K, v) is tame then every maximal immediate extension of (K, v) is tame.*

Remark 3.4. The converse direction of Theorem 3.2 deliberately does not mention whether (K, v) is tame. Tameness of (K, v) requires (TF3), which is not preserved downward through an immediate extension: a proper immediate algebraic extension of (K, v) need not remain immediate after passing to (Z, w) , and conversely immediate algebraic extensions of (Z, w) need not descend. Only (TF1) and (TF2), being statements about two abstract groups, are genuinely symmetric. This is the precise reason the criterion has the shape it does, and the reason the notion of tame base cannot be the right invariant.

4 Decidability transfer

Theorem 4.1 (Relative completeness and decidability for tame fields). *Let (L_1, v_1) and (L_2, v_2) be tame valued fields. Then*

$$(L_1, v_1) \equiv (L_2, v_2) \text{ in } \mathcal{L}_{\text{val}} \iff \Gamma_1 \equiv \Gamma_2 \text{ and } \kappa_1 \equiv \kappa_2.$$

If $\text{char } K = \text{char } \kappa(v)$, then relative completeness and relative decidability both hold [5, Thm. 1]: $\text{Th}(L, v)$ in $\mathcal{L}_{\text{val}}$ is computable from $\text{Th}(\Gamma_v)$ in the language of ordered groups and $\text{Th}(\kappa(v))$ in the language of rings.

The equal-characteristic hypothesis is not cosmetic. In mixed characteristic, tame fields with the same value group and the same residue field need not be $\mathcal{L}_{\text{val}}$ -elementarily equivalent [8, Kuh26, §3], so the *if and only if* is unavailable. Relative decidability in mixed characteristic, on the other hand, is not known to fail but is an open question; Kuhlmann and Lisinski supply additional hypotheses under which equivalence does hold [10, Lisinski, Thm. 2]. This is exactly the gap that leaves mixed-characteristic families such as $\mathbb{Q}_p(p^{1/p^\infty})$ untreated by the transfer below, and it is a gap in the theory rather than in this proof.

Theorem 4.2 (Decidability transfer). *Let (K, v) be a valued field of equal characteristic, $\text{char } K = \text{char } \kappa(v) = p$, such that $\text{Th}(\Gamma_v)$ and $\text{Th}(\kappa(v))$ are recursive. Suppose (K, v) satisfies (TF1) and (TF2). Then every maximal immediate extension (Z, w) of (K, v) has decidable $\mathcal{L}_{\text{val}}$ -theory.*

Proof. By Theorem 3.2, (Z, w) is tame. Since the extension is immediate, $\Gamma_w \cong \Gamma_v$ and $\kappa(w) \cong \kappa(v)$, so the two input theories are unchanged and remain recursive. We are in the equal-characteristic case, so Theorem 4.1 applies and gives decidability of $\text{Th}(Z, w)$. $\square$

Corollary 4.3 (Negative certificate). *Let (K, v) be of equal characteristic with $\text{Th}(\Gamma_v)$ and $\text{Th}(\kappa(v))$ recursive. If some maximal immediate extension (Z, w) of (K, v) has undecidable $\mathcal{L}_{\text{val}}$ -theory, then (K, v) fails (TF1) or (TF2), and hence is not tame.*

Proof. Take the contrapositive of Theorem 4.2. $\square$

Corollary 4.3 is a genuine test rather than a tautology: the hypotheses about Γ_v and $\kappa(v)$ are assumptions about the *base*, while the conclusion is a failure of tameness for the base, extracted from a fact about an extension.

5 The boundary of tameness

Decidability in L_{val} is not the sharpest statement available, and the interesting failures occur one constant at a time.

Proposition 5.1 (Monotonicity under expansion). *Let M' be an expansion of M . If $\text{Th}(M')$ is decidable, then $\text{Th}(M)$ is decidable in the reduct language.*

Proof. Every reduct sentence is a sentence in the richer language. Enumerate reduct sentences and query the decider for $\text{Th}(M')$; validity agrees under expansion. $\square$

Proposition 5.1 fixes the direction of any phenomenon of this shape: the decidable theory must be the *reduct*. A structure whose L_{val} -theory is undecidable while its $L_{\text{val}}(t)$ -theory is decidable cannot exist, for any structure at all. The genuine phenomenon is the reverse.

Theorem 5.2 (Sharpness). *Let*

$$H = \overline{\mathbb{F}}_p((t))^{1/p^\infty} = \bigcup_n \overline{\mathbb{F}}_p((t^{1/p^n})),$$

where $\overline{\mathbb{F}}_p$ is the algebraic closure of the prime field. There are maximal immediate extensions (Z, w) of H such that $\text{Th}(Z, v)$ is decidable in L_{val} and $\text{Th}(Z, t)$ is undecidable in $L_{\text{val}}(t)$.

Proof. Reported result, obtained from the tilting correspondence [9, Kartas], as summarised in [8, Kuh26, §4], and resting on the Fontaine–Wintenberger theorem in its refinements [13]. The tilts in question are maximal immediate extensions of $\overline{\mathbb{F}}_p((t))^{1/p^\infty}$; being tame they are decidable in L_{val} , and they are not decidable in $L_{\text{val}}(t)$. For consistency with the rest of the paper, note independently that the base satisfies (TF1), since its value group $\mathbb{Z}[1/p]$ is p -divisible, and (TF2), since $\overline{\mathbb{F}}_p$ is perfect. Theorem 3.2 therefore also certifies that these fields are tame. $\square$

It is worth noting what the perfect hull itself does *not* satisfy. It is perfect, but it is not algebraically maximal: adjoining a root of the Artin–Schreier polynomial $X^p - X - t^{-1}$ yields an immediate algebraic extension of degree p , so (TF3) fails and the hull is not tame [8, Kuh26, §1]. Tameness appears precisely at the next stage of maximal immediate extension. The hull is the natural place to look for sharpness and yet is not itself sharp.

6 Worked examples

Field	char κ	Γ	(TF1)	(TF2)	Ext. tame?
$\mathbb{F}_p((t))$	p	$\mathbb{Z}$	no	yes	no
$\mathbb{F}_p(t)((t))$	p	$\mathbb{Z}$	no	no	no
$\mathbb{F}_p((t^{1/p^\infty}))$	p	$\mathbb{Z}[1/p]$	yes	yes	yes
$\overline{\mathbb{F}}_p((t^{1/p^\infty}))$	p	$\mathbb{Z}[1/p]$	yes	yes	yes
$\mathbb{Q}_p(p^{1/p^\infty})$	p	$\mathbb{Z}[1/p]$	yes	yes	yes
$\mathbb{Q}_p(\zeta_{p^\infty})$	p	$\mathbb{Z}$	no	yes	no

Four rows deserve comment.

First, $\mathbb{F}_p(t)((t))$ is the one row in which (TF2) fails. Its residue field $\mathbb{F}_p(t)$ is not perfect, since t is not a p -th power in $\mathbb{F}_p(t)$. This is worth separating from the $\mathbb{F}_p((t))$ row, which fails only on (TF1) although its residue $\mathbb{F}_p$ is perfect. A residue field of characteristic zero would instead be perfect automatically, since every field of characteristic zero is perfect, so the obstruction is genuinely a positive-characteristic one.

Second, the equal-characteristic rows above are decidable for a familiar reason: power series fields over finite fields have decidable L_{val} -theory whenever the exponent group is divisible or elementarily equivalent to the p -divisible hull of $\mathbb{Z}$ [6], restated as Theorem 3 of [8], and tame positive-characteristic Hahn fields over a perfect residue are decidable even with a named element [10, Kuh26, Thm. 4].

Third, $\mathbb{Q}_p(p^{1/p^\infty})$ and $\mathbb{Q}_p(\zeta_{p^\infty})$ share residue characteristic and residue field but have different value groups, $\mathbb{Z}[1/p]$ and $\mathbb{Z}$, and the criterion separates them cleanly. Both are mixed characteristic, so Theorem 4.2 does not apply to either; indeed $\mathbb{Q}_p(p^{1/p^\infty})$ admits decidable maximal immediate extensions while $\mathbb{Q}_p(\zeta_{p^\infty})$ has maximal immediate extensions with pairwise distinct L_{val} -theories, so uncountably many of them are not decidable. Value group divisibility, not residue characteristic, is what the criterion tracks.

Fourth, the table lists no decidability column, because decidability is not determined by (TF1) and (TF2) alone: the characteristic and the elementary theories of Γ and κ are independent inputs. This is a feature of the theory, not an incompleteness of the criterion.

7 Applications

7.1 Effective certification of decidability

Theorem 4.2 is constructive in its first three hypotheses. The residue characteristic is computable, p -divisibility of an explicitly presented ordered abelian group is decidable, and perfectness of a residue field is decidable for the field constructions that arise in practice. So one can write a procedure that accepts a description of a valued field and either certifies that every maximal immediate extension has decidable L_{val} -theory or reports that the criterion does not apply. The certification concerns an object that is in general not finitely presentable, which is what makes it worth having.

The four obligations are independent of one another, and conflating the last two is the easiest way to get this wrong. Every field of characteristic zero is perfect, so $\mathbb{Q}$ satisfies the perfection hypothesis and $\mathbb{Q}((t^\mathbb{Q}))$ is genuinely tame. Yet [14] proved that the complete theory of $\mathbb{Q}$ in the language of rings is undecidable, and [15] extended the undecidability to every global field. The residue-field hypothesis therefore fails for the most familiar residue field of characteristic zero, and a certifier that lists $\mathbb{Q}$ among the decidable fields would certify a false statement. What makes this a natural mistake rather than a technical one is that the two recursion hypotheses pull in opposite directions on a single field: the value group $\mathbb{Q}$ is decidable, its residue field is not. Characteristic p exhibits the mirror-image obstruction, and this one is elementary. A rational function field $k(t)$ in characteristic p is never perfect, since $(k(t))^p = k^p(t^p)$ omits t , which is exactly why [10] requires a perfect residue field, and why $\mathbb{F}_p(t)$ cannot serve as the residue of a positive characteristic Hahn field even though $\mathbb{F}_p$ itself can.

One limitation deserves emphasis, since the application is easy to overstate. Deciding whether the complete theory of an ordered abelian group or of a field is recursive is itself undecidable in general, so no procedure settles the last two hypotheses for arbitrary input. What Theorem 4.2

supports is therefore a one-sided procedure and not a decision procedure. Such a certifier reports *certified* when every hypothesis has been independently justified and *not certified* otherwise, and it never reports a theory to be undecidable, having no way to establish that. Each input it cannot analyse is refused rather than guessed at, and a refusal always names the clause that blocked it. That behaviour is what makes the tool useful for surveying families of candidate fields, because a refusal distinguishes a genuinely non-tame base from a base the tool merely failed to analyse.

7.2 Permanence and invariant transport

Corollary 3.3 says tame fields form a class closed under maximal immediate extension. Since tameness underpins relative completeness, relative decidability, the stable embedding theorems for tame fields and the classification of extremal valued fields [11], any such property available for a base is available for its maximal immediate extension at no additional cost. Work on a tame field therefore propagates to an object of very different size and structure, which is often the only tractable representative of the theory.

7.3 Proxy reduction of first-order questions

By Theorem 4.1, two equal-characteristic tame fields with the same value group and residue field are L_{val} -elementarily equivalent. Specialising to a field and its own maximal immediate extension, (Z, w) may be substituted for (K, v) in any L_{val} -question, including questions whose truth value one expects to be easier to determine on the more saturated object. Existential questions about equicharacteristic henselian valued fields are the standard test case here [12]. For equal-characteristic tame fields this substitution is free, since the value group and residue field are literally unchanged. In mixed characteristic the unrestricted form of Theorem 4.1 fails, so the substitution carries no such general guarantee.

7.4 Negative certificates and pruning candidate classes

Corollary 4.3 runs the argument backwards as a screening tool. If an experiment produces a maximal immediate extension with undecidable L_{val} -theory, the base is certified not tame, so the pair is of no further use for tame-field methods and can be set aside. In a survey of candidate fields this is worth having: it replaces an attempt to prove decidability, which may be research-hard, with the much easier task of computing one theory.

7.5 Guiding the choice of base field

Because the criterion reduces the question to (TF1) and (TF2), it tells us which family to work in. Equal characteristic plus a perfect residue field plus a p -divisible value group puts the field in the regime where decidability follows. Characteristic 0 residue with $\mathbb{Z}$ values, as in $\mathbb{F}_p((\zeta_p\text{-free})_{p^\infty})$ style constructions, sits outside it, and one should expect obstruction rather than decidability.

7.6 Bounding where the technique can be pushed

Corollary 4.3 is informative negatively. It shows that the entire equal-characteristic transfer rests on nothing more than two elementary conditions on Γ_v and $\kappa(v)$. Any attempt to extend decidability to a family outside those conditions must supply genuinely new input, not a refinement of Theorem 2.2. That is a useful constraint on the shape of further work.

7.7 Sharpness as a warning about expansion

Proposition 5.1 and Theorem 5.2 together locate the boundary precisely. Decidability of the valued-field theory carries no guarantee about the expansion by a named element, even when the two structures have identical value group and residue field. Applications that name an element should therefore state their result in L_{val} or expect to pay separately.

8 Open problems

1. Whether $\text{Th}(Z, v)$ is decidable for maximal immediate extensions of $\overline{\mathbb{F}}_p((t))$. The base fails (TF1), so the criterion does not apply, and neither does Theorem 4.2.
2. Relative decidability for tame fields in mixed characteristic. Unlike relative completeness, which is known to fail there by the examples following Theorem 4.1, decidability is simply open; Lisinski’s Theorem 2 gives a criterion under extra hypotheses [10, Lis21].
3. Extending Theorem 4.2 to mixed characteristic, which would follow from the preceding item.
4. Decidability of $\mathbb{Q}_p(\zeta_{p^\infty})$, $\mathbb{Q}_p(p^{1/p^\infty})$ and $\mathbb{Q}_p^{\text{ab}}$ themselves, and whether the maximal immediate extensions of the first two can be classified beyond the value-group criterion of Theorem 3.2.

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